

Asmi-Core V1 and Dristi-Net V2: Dual Cascaded Attention Architectures for High-Accuracy Bengali Handwritten Character Recognition

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Abstract

Handwritten Bengali character recognition remains a challenging pattern recognition problem due to large intra-class variability, diverse writing styles, stroke deformations, and visually similar character structures. This paper presents two novel cascaded deep learning architectures, namely **Asmi-Core V1 (AC-V1)** and **Dristi-Net V2 (DN-V2)**, designed for robust Bengali handwritten character classification. Both models employ a two-stage cascaded learning framework combining multi-scale convolutional feature extraction and a structural refinement network. The first architecture, AC-V1, emphasizes channel-wise feature calibration using Squeeze-and-Excitation attention, while the second architecture, DN-V2, extends the framework with a dual-domain attention mechanism integrating both channel and spatial attention. Experimental evaluation demonstrates that AC-V1 achieves **98.87% classification accuracy** using **14,555,300 parameters** after 40 training epochs, whereas DN-V2 utilizes **14,555,496 parameters** and attains comparable high-performance recognition after 45 epochs through enhanced spatial localization capabilities and achieves **98.93% accuracy**. The proposed architectures demonstrate the effectiveness of cascaded refinement and attention-guided feature processing for complex Bengali handwritten character recognition tasks.

Keywords: Bengali Character Recognition, Deep Learning, Attention Mechanisms, Squeeze-and-Excitation, CBAM, Convolutional Neural Networks, Handwritten Character Recognition, Cascaded Learning.

1. Introduction

Optical Character Recognition (OCR) has witnessed significant advancements through deep learning techniques. Despite substantial progress in Latin and Chinese script recognition, Bengali handwritten character recognition remains a challenging research area due to:

- Large character vocabulary.

- High intra-class variation.
- Similar visual structures among distinct characters.
- Stroke distortions and writing inconsistencies.
- Variations in scale, orientation, and handwriting style.

Recent advances in convolutional neural networks (CNNs) have improved recognition performance; however, distinguishing visually similar Bengali characters still requires effective attention mechanisms capable of identifying subtle structural differences.

To address these challenges, this work proposes two novel architectures:

1. **Asmi-Core V1 (AC-V1)** – A channel-attention-driven cascaded architecture.
2. **Dristi-Net V2 (DN-V2)** – A dual-attention cascaded architecture combining channel and spatial attention.

The main objective is to investigate whether spatial attention can further improve discriminative localization beyond channel-wise feature calibration.

2. Related Work

Traditional Bengali OCR systems relied on handcrafted descriptors such as:

- Histogram of Oriented Gradients (HOG)
- Zernike Moments
- Gabor Filters
- Local Binary Patterns (LBP)

The emergence of deep learning introduced CNN-based architectures including:

- LeNet
- AlexNet
- VGG
- ResNet
- DenseNet
- EfficientNet

Attention mechanisms further enhanced feature discrimination through:

- Squeeze-and-Excitation (SE) Networks
- Convolutional Block Attention Module (CBAM)
- Efficient Channel Attention (ECA)
- Coordinate Attention

However, limited research has explored cascaded dual-stage attention architectures specifically optimized for Bengali handwritten character recognition.

3. Proposed Methodology

3.1 Overall Framework

Both architectures follow a two-stage cascaded learning strategy:

Stage 1

Multi-scale feature extraction and preliminary classification.

Stage 2

Feature refinement and structural correction through a dedicated neural refinement module.

The overall workflow is:

Input Image → Multi-Scale Feature Extraction → Attention Processing → Stage-1 Classification
→ Feature Fusion → Structural Refinement → Final Prediction

4. Asmi-Core V1 (AC-V1)

4.1 Architectural Philosophy

The term **Asmi** originates from Sanskrit and signifies self-existence or identity. Accordingly, AC-V1 focuses primarily on internal channel relationships to establish strong feature representations.

4.2 Convolutional Stem

Input image dimensions:

$$1 \times 32 \times 32$$

The stem progressively expands feature representations through stacked convolution layers utilizing Mish activation functions.

Feature progression:

$$1 \rightarrow 32 \rightarrow 64 \rightarrow 128 \text{ channels}$$

4.3 UltraInception Multi-Scale Module

Each UltraInception Block contains four parallel processing paths:

Branch A

1×1 Convolution

Branch B

3×3 Convolution

Branch C

Factorized dual 3×3 convolutions approximating a 5×5 receptive field

Branch D

Max-Pooling followed by projection

Outputs from all branches are concatenated and fused.

4.4 Channel Attention

AC-V1 employs Squeeze-and-Excitation attention.

The attention mechanism performs:

1. Global Average Pooling
2. Channel Compression
3. Nonlinear Transformation
4. Channel Recalibration

This process identifies the most informative feature channels while suppressing redundant activations.

4.5 Cascaded Refinement Stage

The second stage combines:

- Stage-1 feature embeddings (1024 features)
- Auxiliary visual stream (512 features)
- Stage-1 class predictions

These vectors are concatenated and processed by a deep residual MLP.

Residual refinement equation:

$$h_2 = FC_2(h_1) + h_1$$

This enables error correction before final classification.

4.6 Model Statistics

Metric	Value
Parameters	14,555,300
Epochs	40
Accuracy	98.87%

5. Dristi-Net V2 (DN-V2)

5.1 Architectural Philosophy

The word **Dristi** means vision or focused gaze.

Unlike AC-V1, Dristi-Net V2 not only identifies important features but also determines their spatial location within the image.

5.2 Dual Attention Mechanism

The architecture introduces CBAM-inspired sequential attention:

Channel Attention

Determines:

“What is important?”

Spatial Attention

Determines:

“Where is it located?”

This combination enables stronger localization of fine character details.

5.3 Spatial Attention Layer

After channel recalibration, a dedicated spatial attention module is applied.

The module computes:

- Average Pool Map
- Maximum Pool Map

These maps are concatenated and passed through a:

7×7 convolution

to generate a spatial saliency mask.

The resulting mask highlights:

- Dots
- Loops
- Curves
- Accent strokes
- Junction points

which are crucial for Bengali character discrimination.

5.4 Extremal Refiner

The refinement stage receives:

- Intermediate logits
- Spatial embeddings
- Dense visual descriptors

A deep MLP-based correction network resolves classification ambiguities by jointly analyzing structural and visual evidence.

5.5 Model Statistics

Metric	Value
Parameters	14,555,496
Epochs	45
Accuracy	High-Performance Recognition
Extra Parameters over AC-V1	196

The spatial attention module increases the parameter count by only 196 parameters.

6. Experimental Setup

Hardware

Training was performed using deep learning acceleration hardware supporting PyTorch.

Input Resolution

32×32 grayscale images

Optimizer

AdamW

Activation Function

Mish

Loss Function

Cross-Entropy Loss

Training Strategy

- Data augmentation
 - Multi-stage learning
 - Cascaded refinement
 - Attention-guided optimization
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7. Results and Discussion

Comparative Performance

Feature	AC-V1	DN-V2
Attention Type	SE	SE + Spatial
Parameter Count	14,555,300	14,555,496
Epochs	40	45
Multi-Scale Fusion	Yes	Yes
Cascaded Refinement	Yes	Yes
Spatial Localization	No	Yes
Stroke Awareness	Moderate	High
Fine Detail Detection	Moderate	Very High

Analysis

Asmi-Core V1

Advantages:

- Faster convergence.
- Strong channel discrimination.
- Efficient parameter utilization.
- Excellent accuracy.

Suitable when:

- Computational efficiency is important.
 - Training time must be minimized.
-

Dristi-Net V2

Advantages:

- Enhanced spatial localization.
- Better recognition of subtle handwriting variations.
- Improved robustness against stroke distortions.
- Strong geometric awareness.

Suitable when:

- Fine-grained classification is critical.
 - Similar character classes must be distinguished.
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8. Scientific Contributions

The major contributions of this work are:

1. Introduction of the novel Asmi-Core V1 architecture.
 2. Introduction of the novel Dristi-Net V2 architecture.
 3. Development of a cascaded dual-stage recognition framework.
 4. Integration of multi-scale UltraInception modules.
 5. Demonstration of lightweight spatial attention augmentation requiring only 196 additional parameters.
 6. Achievement of 98.87% recognition accuracy using a 14.56M parameter network.
 7. Establishment of an effective architecture for Bengali handwritten character recognition.
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9. Future Work

Future research directions include:

- Vision Transformers (ViT)
 - Swin Transformers
 - ConvNeXt architectures
 - Mamba-based vision models
 - Hybrid CNN-Transformer systems
 - Self-supervised pretraining
 - Multilingual Indic character recognition
 - Edge-device deployment through pruning and quantization
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10. Conclusion

This paper introduced two novel cascaded attention-based architectures for Bengali handwritten character recognition: Asmi-Core V1 and Dristi-Net V2. Both models employ a two-stage learning strategy combining multi-scale feature extraction and structural refinement. AC-V1 focuses on channel-wise feature calibration and achieved 98.87% classification accuracy with 14,555,300 parameters after 40 epochs. DN-V2 extends this framework by incorporating spatial attention, enabling more precise localization of critical handwriting structures while increasing the parameter count by only 196 parameters. Experimental analysis demonstrates that attention-guided cascaded learning provides an effective solution for complex Bengali handwritten character recognition tasks and offers a promising direction for future OCR research in Indic languages.

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Research Areas:

- Deep Learning
- Computer Vision
- Bengali OCR
- Pattern Recognition
- Artificial Intelligence
- Neural Network Architectures

- **Important tables**

Table:1. Comparative Analysis

Compared with previously evaluated architectures:

Model	Parameters	Accuracy
Classical QDA	N/A	74.17%
Super-Charged LeNet	0.66M	95.60%
CNN–Transformer Fusion	14.12M+	93.37%
Fast14M Residual Hybrid CNN	14.47M	98.57%
Proposed Cascaded Pipeline(Cascade-13.61M)	13.61M	98.93%

The proposed framework achieved the highest accuracy among all evaluated systems.

Table 2. Deep Learning Model Comparison

Model	Parameters	Accuracy	F1 Score
Super-Charged LeNet	664,306 (~0.664M)	95.60%	0.9562
Fast14M Residual LeNet Hybrid	14.47M	98.57%	0.99
LeNet → ResNet → VGG → EfficientNet → ViT combined model	14.12M+	93.37%	0.93
**cascade 13.61M	13,606,756	98.93%	-

Model	Parameters	Accuracy	F1 Score
ASMI-CORE V1	14,555,300	98.87%	-
DRISTI-NET V2 (DN-V2)	14,555,496	98.93%	-

Table 3. Accuracy Gain Relative to Classical ML

Model	Accuracy
Best Classical ML (QDA)	74.17%
Super-Charged LeNet	95.60%
Fast14M Hybrid CNN	98.57%
**cascade 13.61M	98.93%
ASMI-CORE V1	98.87%
DRISTI-NET V2 (DN-V2)	98.93%

Table 4. Classical AutoML Classifier Performance

#	Model	Accuracy	F1 Score	Precision	Recall	Train Time	Predict Time
1	Quadratic Discriminant Analysis	74.17%	0.7426	0.7969	0.7417	1.0171s	0.2508s

#	Model	Accuracy	F1 Score	Precision	Recall	Train Time	Predict Time
2	Support Vector Classifier (SVC)	73.97%	0.7443	0.7633	0.7397	15.2699s	13.8084s
3	K-Neighbors (K=5)	68.60%	0.6829	0.7071	0.6860	0.0224s	1.7526s
4	K-Neighbors (K=3)	67.97%	0.6786	0.7005	0.6797	0.0218s	2.0042s
5	Multi-Layer Perceptron (MLP)	62.87%	0.6271	0.6294	0.6287	4.3618s	0.0068s
6	Linear SVC	51.97%	0.5157	0.5160	0.5197	19.1463s	0.0200s
7	Gaussian Naive Bayes	51.10%	0.5009	0.5403	0.5110	0.0342s	0.1282s
8	Linear Discriminant Analysis	50.50%	0.5069	0.5167	0.5050	0.3038s	0.0299s
9	SGD Classifier	49.47%	0.4969	0.5121	0.4947	9.3716s	0.0099s
10	Random Forest	49.40%	0.4865	0.4918	0.4940	15.8854s	0.5770s
11	Nearest	46.37%	0.4587	0.4839	0.4637	0.0599	0.0544s

#	Model	Accuracy	F1 Score	Precision	Recall	Train Time	Predict Time
	Centroid					s	
12	Ridge Classifier	45.57%	0.4396	0.4698	0.4557	0.1494s	0.0127s
13	Extra Trees	42.00%	0.4092	0.4099	0.4200	2.3407s	0.5910s
14	Perceptron	41.90%	0.4072	0.4182	0.4190	3.5103s	0.0092s
15	Bagging Classifier	41.87%	0.4161	0.4331	0.4187	62.7010s	0.0276s
16	Bernoulli Naive Bayes	41.13%	0.4113	0.4161	0.4113	0.1013s	0.0287s
17	Passive Aggressive	38.67%	0.3875	0.4154	0.3867	5.7374s	0.0167s
18	Decision Tree	30.47%	0.3033	0.3062	0.3047	11.8614s	0.0033s
19	Extra Tree	12.43%	0.1241	0.1252	0.1243	0.1503s	0.0037s
20	AdaBoost	6.47%	0.0151	0.0093	0.0647	26.9335s	0.0390s
21	Dummy Baseline	2.00%	0.0008	0.0004	0.0200	0.0028s	0.0003s

Table 5. PCA-Based AutoML Benchmark Results

Classifier Name	Accuracy	Train Time (s)
Logistic Regression	57.67%	20.3854s
Ridge Classifier	45.57%	0.1729s
SGD Classifier	49.87%	14.4397s
Perceptron	41.83%	5.2316s
Passive Aggressive	39.50%	8.5623s
Linear SVC	51.97%	25.9259s
Support Vector Classifier (SVC)	73.97%	16.8608s
K-Neighbors (K=3)	67.97%	0.0029s
K-Neighbors (K=5)	68.60%	0.0048s
Nearest Centroid	46.37%	0.0548s
Decision Tree	31.37%	15.3191s
Extra Tree	11.87%	0.1044s
Random Forest	58.00%	71.5330s
Extra Trees	54.33%	11.8140s
AdaBoost	8.80%	62.5106s

Table 6. Modified extremely lightweight versions with Overall Accuracy and Parameter Counts(trying to limit with 200k-300k parameters)

cascade mini lightweight models	Overall Accuracy	Parameters	Dashboard Title
cascade-mini-1	91.17%	121,796 / 200,000	Cascaded Optimization System Dashboard
cascade-mini-2	96.80%	180,684 / 200,000	Cascaded Optimization System Dashboard
cascade-mini-3	96.97%	236,372 / 200,000	Cascaded System Optimization Dashboard
cascade-mini-4	97.33%	314,100	Cascaded System Optimization Dashboard
**cascade 13.61M	98.93%	13,606,756	Cascaded System Optimization Dashboard
ASMI-CORE V1	98.87%	14,555,300	Cascaded System Optimization Dashboard
DRISTI-NET V2 (DN-V2)	98.93%	14,555,496	Cascaded System Optimization Dashboard